Research Article

Stem Cell Growth using Mycelium Framework

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Abstract

Mycelium

Keywords: Place text here

1. INTRODUCTION

Using myclium scaffold to influence stem cell adaptation. Creating the stem cell niche by growing the stem cell in a natural biological scaffold.

2. LITERATURE REVIEW

In [1], the affects of a biological gause on wound healing is observed. It concludes that the mycelium biomaterials are a promising approach to alternative accelrative wound healing. Both tested materials, G. lucidum and P. ostreatus enhance collagen I gene expression. G. lucidum performs marginally better. The β-glucans in the mycelia are crucial in prompting the body to synthesize types I and III collagen.

In [2] it is demonstrated that the mycelia of Pleurotus ostreatus can be directly used as the scaffold for cell growth. It had good viability, and morphology compared to the control samples used in [2]. Human cells were directly plated onto the un-moddified and non-functionalised mycelial scaffold. Worth noting; even after Hyphal cell deactivation, Ganoderma lucidum was not fully compatible for direct plating. Its water extract contained concentrations of organic acids (Ganoderic acid V and Oleandrin). This was detrimental for HDFa cells survivial

In [3], the stem cell niche is discussed. It details the effect of the microenvironment on stem cell differenciation. It provides a methodology for defining the cell types that stem cells can grow into, and the condiitons that are needed for this to occur. Under homeostatic conditions, there is a symbiotic relationship between the stem cells and their niche. What must be further investigated is the response of the stem cells and their niche to tissue injury during the disease progression and aging. Further study must be done on genetic factors that influence the stem cell niche.

3. METHODOLOGY AND DATA

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3.1. Example Subsection

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3.2. Example Second Subsection

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3.2.1. Example Subsubsection

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4. RESULTS AND DISCUSSION

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5. CONCLUSION AND FUTURE WORK

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6. DECLARATION

Conflict of Interest: Place text here

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7. REFERENCES

[1] M. Ruggeri, D. Miele, M. Contardi, et al. "Mycelium-based biomaterials as smart devices for skin wound healing". In: *Frontiers in Bioengineering and Biotechnology* 11 (2023), p. 1225722. DOI: 10.3389/fbioe.2023.1225722. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC10465301/.

[2] Maria Elena Antinori et al. "Advanced mycelium materials as potential self-growing biomedical scaffolds". In: *Scientific Reports* 11.1 (2021), p. 12630. DOI: 10.1038/s41598-021-91572-x. URL: https://doi.org/10.1038/s41598-021-91572-x.

[3] Justin Voog and D. Leanne Jones. "Stem Cells and the Niche: A Dynamic Duo". In: *Cell Stem Cell* 6.2 (2010), pp. 103–115. DOI: 10.1016/j.stem.2010.01.011. URL: https://doi.org/10.1016/j.stem.2010.01.011.